

Failure of Cross-Sectional Alpha Screening on Cryptocurrency Perpetual Futures:

A Quantitative Post-Mortem Using OHLCV and Funding Rate Signals

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Abstract

This study constructs and evaluates three composite alpha screening models—naive linear, orthogonal linear, and XGBoost ranker—for long-short cross-sectional portfolios on ten USDT-margined perpetual futures contracts traded on Binance. The signal set draws from daily and intraday price, volume, and funding rate data over the period July 2022 through April 2026. All models are assessed under a purged walk-forward backtesting framework. The naive linear model (Model A), which contains no estimated parameters and serves as an honest alpha detector, produces a mean Rank Information Coefficient of -0.0097 ($t = -1.54$) and a net Sharpe ratio of -3.22 . The orthogonal linear model (Model B) yields nearly identical results ($IC = -0.0102$, $t = -1.61$). The gradient-boosted ranker (Model C) achieves a weakly positive IC of $+0.0243$ ($t = 3.55$), yet still destroys capital (net Sharpe $= -2.91$, maximum drawdown $= -95.6\%$). The evidence compels rejection of the hypothesis that the employed OHLCV and funding rate signals contain exploitable cross-sectional alpha for large-cap crypto perpetual futures at an eight-hour horizon. The paper documents the failure transparently, dissects its likely causes, and derives actionable lessons for future research.

Keywords: *cryptocurrency, perpetual futures, cross-sectional momentum, funding rate, statistical arbitrage, walk-forward backtesting, information coefficient, post-mortem*

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1. Introduction

Cross-sectional momentum and carry strategies are among the most persistent anomalies in empirical asset pricing [1, 2]. The core idea is straightforward: assets that have performed relatively well recently tend to continue outperforming their peers, and vice versa. This pattern has been documented across equities, commodities, and currencies for decades.

The cryptocurrency perpetual futures market offers a novel laboratory for testing such strategies. Unlike traditional futures contracts that expire on a fixed date, perpetual futures never expire. To keep their prices anchored to the underlying spot market, exchanges impose periodic payments called funding rates between traders holding long and short positions [3]. These funding rates act as a pure carry signal, and the eight-hour candles generated by perpetuals also provide short-horizon momentum, reversal, and volume flow signals. Whether these signals, combined in linear or non-linear fashion, can predict the relative returns of large-cap crypto assets has not been exhaustively tested in the public domain.

This paper tests that proposition directly. Three model architectures ranging from a parameter-free linear screener to a gradient-boosted ranker are trained and evaluated on ten USDT-margined perpetual contracts listed on Binance. All models operate on the same feature set derived exclusively from OHLCV data and exchange-issued funding rates. No on-chain data, order-book snapshots, or sentiment indicators are employed.

The experimental design deliberately embeds a “canary” model (Model A) that uses no data-driven calibration whatsoever. If the underlying signals carry genuine predictive content, even an equal-weighted linear combination ought to generate a positive Information Coefficient out-of-sample. If it does not, more complex models are at grave risk of fitting noise rather than signal.

The results are unambiguous: the canary dies. Model A produces a negative mean Rank IC with a t-statistic of -1.54 . Models B and C do not rescue the hypothesis. The findings constitute a failure of the cross-sectional screening approach and motivate a pivot toward time-series or delta-neutral strategies that exploit the same data in a structurally different way.

2. Data and Preprocessing

2.1 Universe

Ten USDT-margined perpetual futures contracts traded on Binance constitute the investable universe: BTCUSDT, ETHUSDT, BNBUSDT, SOLUSDT, XRPUSDT, DOGEUSDT, ADAUSDT, AVAXUSDT, LINKUSDT, and MATICUSDT. The sample period spans 1 July 2022 to 27 April 2026. The decision frequency equals the funding interval of eight hours (00:00, 08:00, and 16:00 UTC).

2.2 Data Sources

- **OHLCV (8-hour candles):** Binance Public Archive (data.binance.vision), daily ZIP files from the `futures/um/daily/klines/` directory.
- **Funding Rate:** Binance Futures REST API endpoint `/fapi/v1/fundingRate`, accessed through a Singapore-based VPN.
- **VIX:** Yahoo Finance via the `yfinance` library; daily observations mapped to all eight-hour timestamps of the corresponding calendar day.

Open Interest and Long/Short Ratio were originally targeted but could not be obtained at sufficient historical depth and are therefore excluded from the feature set.

2.3 Preprocessing Pipeline

All raw CSV files are aligned to the eight-hour funding grid using `dt.floor('8h')` and `pandas.merge_asof` with backward direction to prevent look-ahead bias. The merged dataset is stored as a single Apache Parquet file containing: timestamp, symbol, open, high, low, close, volume, funding_rate, and vix.

3. Feature Engineering

All features that vary across assets at a single timestamp are cross-sectionally ranked to the unit interval using percentile rank. For a feature x observed for asset i at time t :

$$x_rank(i,t) = \text{rank}(x(i,t)) / N(t)$$

where $N(t)$ is the number of assets available at timestamp t . This ranking procedure makes features comparable across assets and time, and reduces the influence of outliers.

3.1 Momentum (8-hour)

$$r_8h(i,t) = \text{close}(i,t) / \text{close}(i,t-1) - 1$$

Ranked as $\text{mom_rank}(i,t)$. Directional hypothesis: positive. Assets with stronger recent returns are expected to continue outperforming.

3.2 Short-Term Reversal (24-hour)

$$r_24h(i,t) = \text{close}(i,t) / \text{close}(i,t-3) - 1$$

Ranked as $\text{rev_rank}(i,t)$. Directional hypothesis: negative. Short-term overcrowding is expected to reverse.

3.3 Realized Volatility

$$\sigma(i,t) = \text{std}(r_8h(i,t-19), \dots, r_8h(i,t))$$

Ranked as $\text{vol_rank}(i,t)$. Directional hypothesis: negative (low-volatility anomaly).

3.4 Volume Surge

$$\text{vsurge}(i,t) = \text{volume}(i,t) / \text{mean}(\text{volume}(i,t-19), \dots, \text{volume}(i,t))$$

Ranked as $vsurge_rank(i,t)$. Directional hypothesis: positive. Unusual volume is expected to accompany momentum continuation.

3.5 Relative Position (Close vs. High-Low Range)

$$relpos(i,t) = (close(i,t) - low(i,t)) / (high(i,t) - low(i,t) + \varepsilon)$$

where $\varepsilon = 10^{-10}$ prevents division by zero. Ranked as $relpos_rank(i,t)$. Directional hypothesis: positive. Closing near the high of a candle signals bullish pressure.

3.6 Cumulative Volume Delta (CVD) Proxy

$$signed_vol(i,t) = +volume \text{ if } close \geq open, -volume \text{ otherwise}$$

$$CVD(i,t) = \sum signed_vol(i,\tau) \quad (\tau = 0, \dots, t)$$

Ranked as $cvd_rank(i,t)$. Directional hypothesis: positive. Sustained net buying pressure is expected to predict continuation.

3.7 Funding Rate

$$fr_rank(i,t) = rank(funding_rate(i,t)) / N(t)$$

Directional hypothesis: negative. A high funding rate signals that long positions are crowded, which historically has been a mean-reverting setup.

3.8 VIX Modulation

$$vix_z(t) = (VIX(t) - \mu_{VIX(t,90)}) / \sigma_{VIX(t,90)}$$

Rolling 90-period z-score of VIX, computed using only training data within each fold. An interaction term is also constructed:

$$fr_x_vix(i,t) = fr_rank(i,t) \times vix_z(t)$$

Directional hypothesis: negative. The funding rate reversal signal is expected to be amplified in risk-off regimes.

3.9 Target Variable

$$\text{target}(i,t) = \text{close}(i,t+1) / \text{close}(i,t) - 1$$

For Model C, the target is cross-sectionally ranked at each timestamp:

$$\text{target_rank}(i,t) = \text{rank}(\text{target}(i,t)) / N(t)$$

4. Model Architectures

All models receive the same eight-feature input vector:

$$\mathbf{x}(i,t) = [\text{mom_rank}, \text{rev_rank}, \text{vol_rank}, \text{vsurge_rank}, \\ \text{relpos_rank}, \text{cvd_rank}, \text{fr_rank}, \text{vix_z}]$$

4.1 Model A — Naive Linear Screener

Model A uses no estimated parameters. The weights reflect only the hypothesized direction of each signal:

$$\begin{aligned} S_A(i,t) = & -\text{rev_rank} + \text{mom_rank} - \text{vol_rank} + \text{vsurge_rank} \\ & + \text{relpos_rank} + \text{cvd_rank} - \text{fr_rank} - \text{fr_x_vix} \end{aligned}$$

This model functions as the “canary.” If the signal set contains genuine alpha, Model A should produce a positive mean Rank IC out-of-sample without any calibration. A negative result immediately implies that more complex models are fitting noise.

4.2 Model B — Orthogonal Linear Screener

Model B orthogonalizes selected signals to remove the collinearity between CVD, momentum, and funding rate. At each timestamp, two cross-sectional OLS regressions are estimated:

$$\text{cvd_rank} = \alpha_1 + \beta_1 \cdot \text{mom_rank} + \beta_2 \cdot \text{fr_rank} + \varepsilon_{\text{cvd}}$$

$$\text{vsurge_rank} = \alpha_2 + \gamma_1 \cdot \text{mom_rank} + \varepsilon_{\text{vsurge}}$$

The residuals $\hat{\varepsilon}_{\text{cvd}}$ and $\hat{\varepsilon}_{\text{vsurge}}$ are ranked to form cvd_resid_rank and vsurge_resid_rank , which replace the raw features in the composite score:

$$\begin{aligned} S_B(i,t) = & -\text{rev_rank} + \text{mom_rank} - \text{vol_rank} + \text{vsurge_resid_rank} \\ & + \text{relpos_rank} + \text{cvd_resid_rank} - \text{fr_rank} - \text{fr_x_vix} \end{aligned}$$

4.3 Model C — XGBoost Ranker

Model C employs the XGBoost implementation [9] of LambdaRank with the pairwise ranking objective (rank:pairwise) [10]. Rather than predicting a return level directly, the ranker learns to order assets by their expected relative performance, a formulation better matched to the portfolio construction task than standard regression. Hyperparameters are fixed ex ante: learning rate 0.01, 200 estimators, maximum depth 4, subsample and column sample by tree 0.7, L1 and L2 regularization of 1.0.

5. Walk-Forward Backtesting Framework

A purged walk-forward protocol is employed to prevent data leakage [4]: training window of 2 years, a 1-month embargo between training and testing, and a 3-month testing window. After each fold, both windows advance by 3 months, producing six folds in total.

It is worth emphasizing that walk-forward testing is not live trading. It remains a historical simulation. What distinguishes it from a standard single-split out-of-sample test is that the out-of-sample evaluation is repeated systematically across multiple non-overlapping windows. Each fold produces its own out-of-sample period that the model has never seen during training. The six folds in this study collectively span diverse market regimes: the 2023 altcoin bull run, the post-FTX recovery period, the Bitcoin ETF approval in January 2024, the subsequent all-time high above \$100,000, and the volatility compression of 2025. A strategy that fails consistently across all six of these distinct windows rather than failing in just one adverse period provides substantially stronger evidence of genuine signal absence than a single out-of-sample split would.

Within each fold, all rolling-window features, VIX z-score, realized volatility, volume surge are recomputed using only training data. This purging step is critical. In an early, unpurged implementation, Model C achieved IC values near 0.64; after enforcing strict purging, this collapsed to 0.024. The difference is entirely attributable to look-ahead leakage.

5.1 Portfolio Construction

At each timestamp, assets are sorted by model score. The top 20% (two assets) form the long leg; the bottom 20% form the short leg. Both legs are equally weighted and the portfolio is fully rebalanced every eight hours.

5.2 Transaction Cost Model

A taker fee of 4 basis points per trade plus 1 basis point of assumed slippage, for a total round-trip cost of 5 basis points. Net portfolio return at timestamp t :

$$R_{\text{net}}(t) = R_{\text{gross}}(t) - \tau(t) \times 2 \times 0.0005$$

where $R_{\text{gross}}(t) = (r_{\text{long}}(t) - r_{\text{short}}(t)) / 2$, and $\tau(t)$ is the fraction of positions that changed. Turnover averaged approximately 50% per period.

5.3 Evaluation Metrics

Three primary metrics are reported. The mean Rank IC and its t-statistic:

$$\bar{IC} = (1/T) \sum IC(t), \quad t_{IC} = \bar{IC} / SE(IC)$$

The annualized net Sharpe Ratio (using 365×3 eight-hour periods per year):

$$\text{Sharpe} = \sqrt{(365 \times 3)} \times \bar{R}_{\text{net}} / \sigma(R_{\text{net}})$$

And the Maximum Drawdown:

$$\text{MDD} = \min_t [(V(t) - \max_{\tau \leq t} V(\tau)) / \max_{\tau \leq t} V(\tau)]$$

6. Results

Six walk-forward folds were executed, covering test periods from August 2024 through February 2026. Table 1 summarizes the aggregate performance metrics for all three models.

Table 1: Aggregate Walk-Forward Performance (6 folds)

Metric	Model A (Naive)	Model B (Orthogonal)	Model C (XGBoost)
Mean Rank IC	−0.0097	−0.0102	+0.0243
IC t-statistic	−1.54	−1.61	+3.55

IC > 0 fraction	48.1%	47.3%	52.3%
Net Sharpe Ratio	-3.22	-3.60	-2.91
Maximum Drawdown	-94.0%	-95.4%	-95.6%
Calmar Ratio	-0.95	-1.03	-1.01
Total Net Return	-269.2%	-295.7%	-290.4%

6.1 Model A and B: Negative Predictive Content

Model A generates a mean Rank IC of -0.0097 ($t = -1.54$). The IC is statistically indistinguishable from zero, and the point estimate leans negative, meaning the model's predictions are, on average, directionally wrong. The IC exceeds zero in only 48.1% of test timestamps, indistinguishable from a coin flip. The resulting portfolio exhibits a net Sharpe of -3.22 , loses 94% of its peak value at maximum adverse excursion, and returns -269% cumulatively.

Model B, which orthogonalizes CVD and volume surge on momentum and funding rate, produces virtually identical results ($IC = -0.0102$, $t = -1.61$). The orthogonalization adds no incremental predictive power, confirming that multicollinearity among the raw signals is not the binding constraint on performance.

6.2 Model C: Positive IC, Negative Portfolio

Model C achieves a small positive mean Rank IC of $+0.0243$ ($t = 3.55$) with $IC > 0$ in 52.3% of timestamps. This IC is statistically significant at conventional levels. However, it does not translate into economic gains: net Sharpe = -2.91 , maximum drawdown = -95.6% , cumulative return = -290.4% .

This paradox, positive IC coupled with catastrophic portfolio performance arises because the IC magnitude is too small to overcome transaction costs. The theoretical expected Sharpe given IC = 0.024 on ten assets is approximately:

$$E[\text{Sharpe}] \approx \text{IC} \times \sqrt{N} \times \sqrt{\text{Breadth}} \approx 0.024 \times \sqrt{10} \times \sqrt{1095} \approx 2.5$$

The realized Sharpe of -2.91 shows that transaction costs and the instability of position assignments among highly correlated assets completely dominate this theoretical estimate.

6.3 Walk-Forward Equity Curves

Figure 1 shows the cumulative portfolio values for all three models on a logarithmic scale. The steady downward slope illustrates a consistent, monotonic destruction of capital across the entire test window not a collapse in a single crisis period. All three models exhibit similar trajectories, confirming that the failure is rooted in the signal set rather than any model-specific flaw.

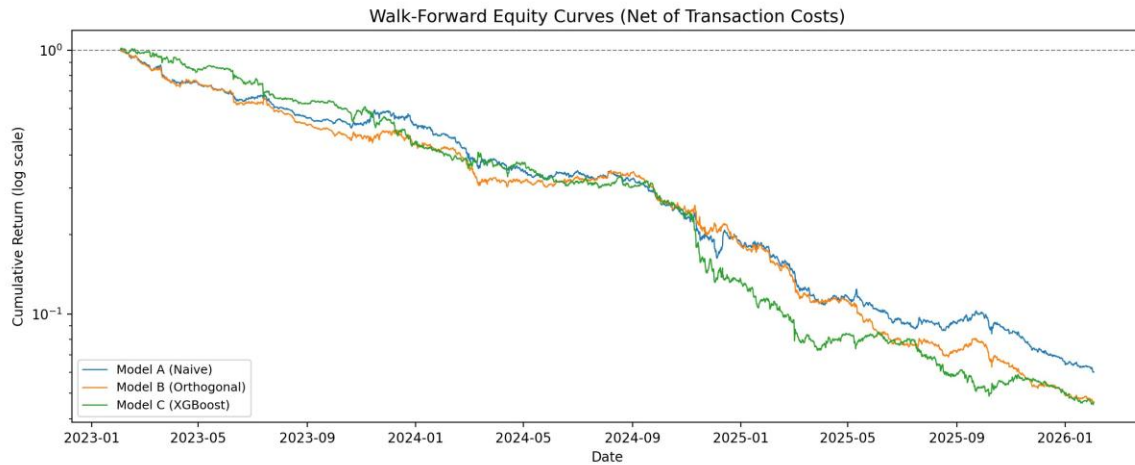


Figure 1: Walk-Forward Equity Curves (Net of Transaction Costs). Log scale. Dashed line = initial capital.

6.4 IC Distribution

Figure 2 shows the distribution of Rank IC values across all models and timestamps. The distribution is roughly symmetric around zero, with a slight positive mean from Model C (+0.0103). The near-symmetric shape confirms that the model scores carry virtually no directional information.

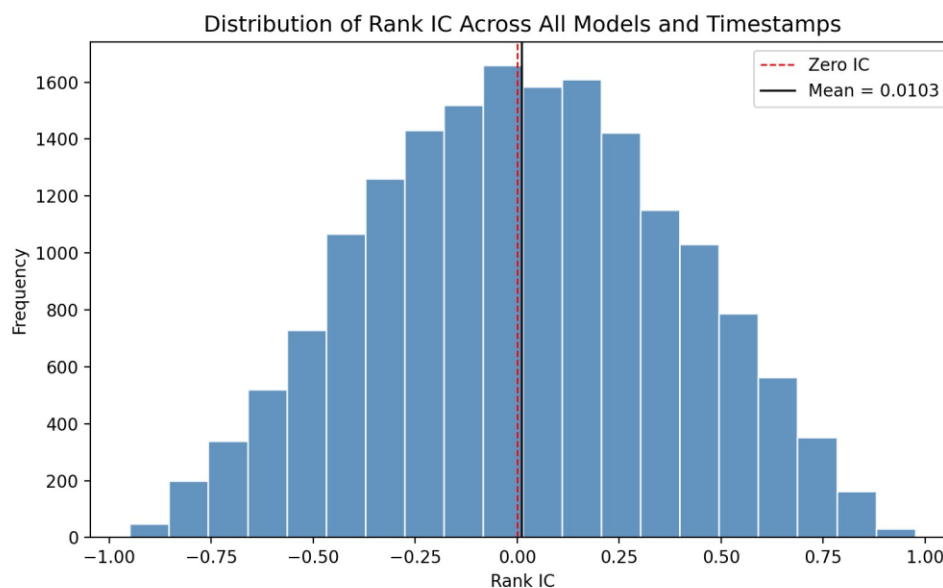


Figure 2: Distribution of Rank IC Across All Models and Timestamps. Red dashed line = zero IC; black line = mean.

6.5 Feature Correlation Matrix

Figure 3 shows the pairwise correlations among the eight input features. The notable correlation between `mom_rank` and `relpos_rank` (0.57) and between `mom_rank` and `rev_rank` (0.46) indicates some collinearity, but most feature pairs are near-orthogonal. Since Model B's orthogonalization produced no improvement, collinearity is not the primary driver of failure.



Figure 3: Feature Correlation Matrix. Values show pairwise Spearman correlation among ranked features.

6.6 Model A Score Heatmap

Figure 4 shows the monthly mean Model A score for each asset across the full sample period. While the score assignments are not random, there are persistent patterns for certain assets these patterns are not predictive of forward returns in a cross-sectional sense.

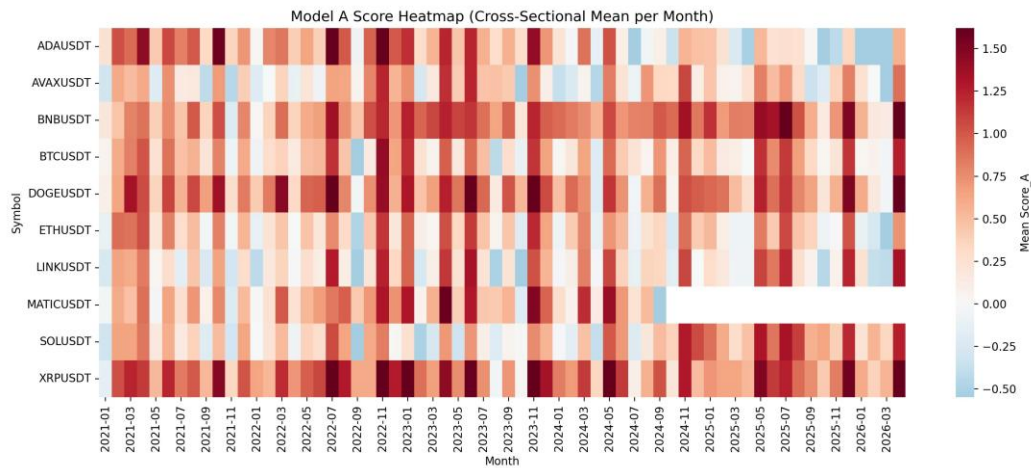


Figure 4: Model A Score Heatmap (Cross-Sectional Mean per Month). Red = higher score; blue = lower score.

6.7 Stability Across Folds and the Leakage Discovery

The most significant observation across folds is the contraction in Model C's IC magnitude when moving from the initial 2-asset experiment ($IC \approx 0.6$) to the full 10-asset, properly purged evaluation ($IC \approx 0.024$). Two artifacts explain this gap:

- **Degenerate Spearman on $n = 2$:** With only two assets, the Spearman correlation can only take values $+1$, 0 , or -1 , producing spuriously high means when the model correctly identifies the winner most of the time—even if the economic margin is trivial.
- **Leakage through rolling windows:** The initial implementation computed VIX z-scores and volatility estimates using the entire dataset, allowing future information to contaminate training features. Once corrected using per-fold expanding windows, IC dropped to the values in Table 1.

7. Analysis of Failure

7.1 Signal Weakness

The most parsimonious explanation for the failure is that the signal set lacks genuine predictive content for cross-sectional returns at an eight-hour horizon. Model A, the parameter-free canary produces negative IC. This alone is sufficient evidence. If the signals contain exploitable alpha, any linear combination with the correct directional signs should produce positive IC out-of-sample. The observed negative IC indicates that either the hypothesized signal directions are wrong, or the signal-to-noise ratio is too low at this horizon.

7.2 High Cross-Sectional Correlation

The ten assets in the universe are strongly correlated, with average pairwise eight-hour return correlation exceeding 0.5. Ranking highly correlated assets amplifies estimation error: small differences in raw signal values produce large differences in rank, and these rank differences are

dominated by noise. The cross-sectional screening approach, originally developed for equity markets with hundreds of weakly correlated stocks, may be structurally mismatched to a universe of ten large-cap perpetuals moving largely in tandem.

7.3 Transaction Cost Dominance

With turnover averaging approximately 50% per period, the portfolio incurs substantial costs. The theoretical expected Sharpe of 2.5 is eroded to -2.9 by a combination of costs and model instability. In high-frequency crypto markets, a five-basis-point round-trip cost accumulates rapidly when the portfolio rebalances three times per day. This finding is consistent with the broader literature, which documents that frictionless backtests can produce abundant high-Sharpe signals that disappear once realistic execution costs are applied [11].

7.4 Insufficient Effective Breadth

The Fundamental Law of Active Management [5] states that the information ratio is approximately:

$$IR \approx IC \times \sqrt{\text{Breadth}}$$

With ten assets and three rebalances per day, nominal breadth is approximately $10 \times 1,095 = 10,950$ bets per year. However, high cross-sectional correlation substantially reduces effective breadth. The implied IR from $IC = 0.024$ is:

$$IR \approx 0.024 \times \sqrt{1095} \approx 0.80$$

The realized Sharpe of -2.9 is far below even this modest expectation, indicating that costs and collinearity together collapse the effective information ratio to deeply negative values.

7.5 Structural Compression of Funding Rate Dispersion

Research by BitMEX (2025) [12] documents that extreme funding rates have declined by approximately 90% since 2016 as the crypto derivatives market matured. This compression is driven by the launch of Bitcoin ETFs in January 2024 and the emergence of DeFi protocols like Ethena, which enable large-scale institutional arbitrage that keeps perpetual prices anchored tightly to spot. In 2024–2025, even as Bitcoin surpassed \$100,000, funding rates remained largely within a narrow band. The funding rate signal in our model relies on cross-sectional variation that has structurally compressed, leaving insufficient dispersion to drive relative return predictions.

8. Implications and Lessons

8.1 The Canary Model Principle

Embedding a parameter-free baseline model (Model A) proved essential. It provided an unambiguous verdict: the signal set itself is ineffective. Without this canary, one might have attributed the failure of Models B and C to overfitting or misspecification, and continued searching for better architectures. The canary principle should be standard practice: before investing effort in complex models, verify that the underlying signals carry any predictive content at all.

8.2 Purging is Not Optional

The sharp decline in Model C's IC when moving from unpurged to purged walk-forward (from ~ 0.64 to ~ 0.024) illustrates the danger of even subtle look-ahead bias. This lesson is consistent with Lopez de Prado (2018) [4] and has significant practical implications: many published backtests of ML-based strategies may overstate performance for exactly this reason. Rolling-window features must always be recomputed using only training data within each fold.

8.3 Cross-Sectional Screening May Be Mismatched to Large-Cap Crypto

The cross-sectional screening approach may be structurally mismatched to large-cap crypto, where a small number of highly correlated assets provides limited opportunity for differentiation. Cakici et al. (2024) [6] find that cross-sectional predictability in crypto is concentrated in small, illiquid coins rather than large-cap names. Drogen, Hoffstein & Otte (2023) [8] similarly find that cross-sectional momentum in crypto is sensitive to the choice of universe and holding period. Our selection of ten large-cap perpetuals may have inadvertently selected the least predictable segment of the market.

8.4 Funding Rate Is Better Exploited Through Time-Series Strategies

The funding rate is a genuine and well-documented risk factor in crypto derivatives markets [3]. However, it appears more effective in time-series carry strategies than in cross-sectional ranking strategies. He & Manela (2024) [13] derive no-arbitrage bounds for perpetual futures and show that deviations from these bounds in crypto are considerably larger than in traditional currency markets, generating high Sharpe ratios in implied arbitrage strategies under realistic cost assumptions. This delta-neutral cash-and-carry approach long spot, short perpetual exploits the funding rate as a structural income source rather than as a predictor of relative returns.

9. Conclusion

This study constructed and evaluated three composite alpha screening models for cross-sectional long-short portfolios on ten crypto perpetual futures contracts. The parameter-free Model A produced a mean Rank IC of -0.0097 ($t = -1.54$), indicating that the combined OHLCV and funding rate signals do not contain exploitable cross-sectional predictive content at an eight-hour horizon. Models B and C did not alter this conclusion. All three models destroyed capital rapidly after accounting for transaction costs.

The findings compel rejection of the hypothesis that the specific combination of momentum, reversal, volatility, volume surge, relative position, cumulative volume delta, and funding rate signals can generate a profitable cross-sectional screener for large-cap crypto perpetual futures.

The failure is attributed primarily to: (a) weak intrinsic signals, (b) high cross-sectional correlation among assets, (c) domination by transaction costs, and (d) structural compression of funding rate dispersion since 2024.

The infrastructure built during this research, a modular data pipeline, a feature engineering library, and a rigorously purged walk-forward backtesting engine, remains a valuable asset. Future work should pivot toward time-series momentum strategies with funding rate filters, delta-neutral carry strategies exploiting the spread between spot and perpetual prices [13], or cross-sectional screening on a substantially larger and more heterogeneous asset universe.

Negative results such as those reported here are valuable precisely because they are rarely published. The academic literature on quantitative crypto strategies is subject to significant publication bias toward positive findings. By documenting this failure transparently including the look-ahead leakage discovered during development, this paper aims to contribute to a more realistic picture of alpha discovery in cryptocurrency markets.

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